Chapter 18 Review

Campbell Biology, AP Edition, 12e

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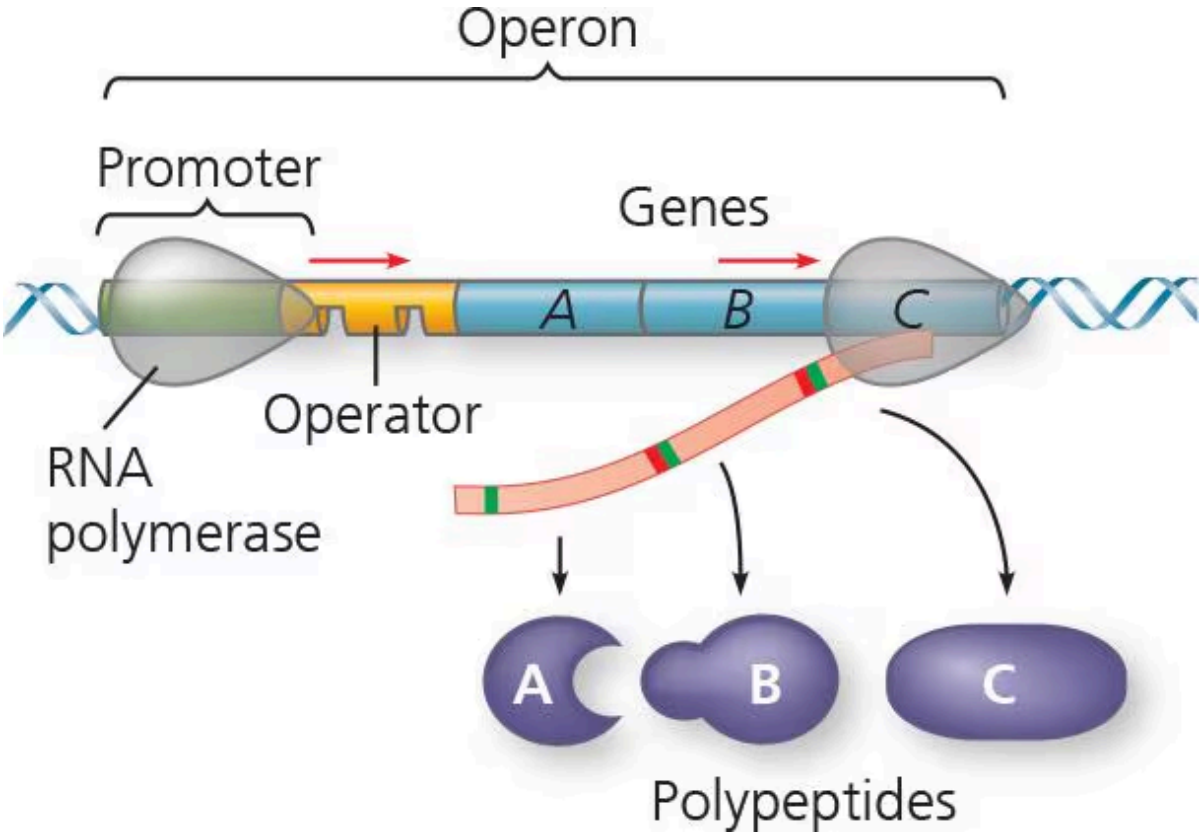
Summary of Key Concepts

In situ hybridization using fluorescent tags shows that certain probes built from *Drosophila* genes label not only specific locations within *Drosophila* embryos, but also similar locations in the embryos of many species of arthropods. Explain this result. **(BIG IDEA 1)**

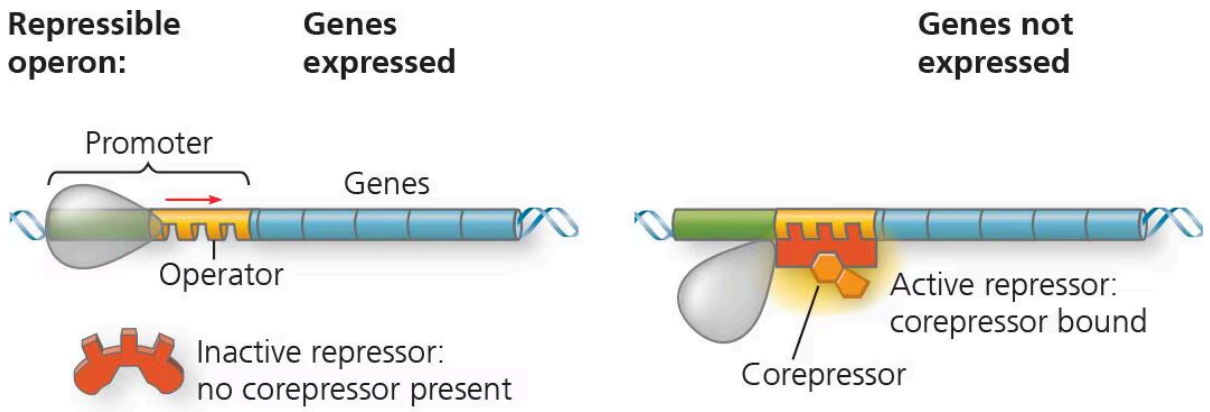
Concept 18.1: Bacteria often respond to environmental change by regulating transcription

- Cells control metabolism by regulating enzyme activity or the expression of genes coding for enzymes. In bacteria, genes are often clustered into **operons**, with one promoter serving several adjacent genes. An **operator** site on the DNA switches the operon on or off, resulting in coordinate regulation of the genes.



Operon diagrammed left to right: D N A with promoter with R N A polymerase bound; followed by operator, genes (A, B, C) being transcribed into continuous m R N A transcript. Transcript has start and stop codons for each gene. Polypeptides A, B and C are translated from the m R N A transcript for each gene.

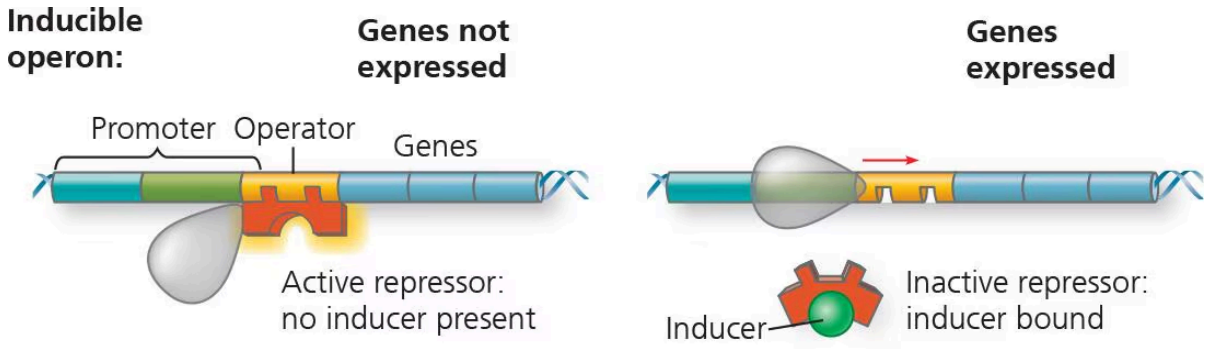
- Both repressible and inducible operons are examples of negative gene regulation. In either type of operon, binding of a specific ****__repressor__ protein to the operator shuts off transcription. (The repressor is encoded by a separate **regulatory gene**.) In a repressible operon (usually encoding anabolic enzymes), the repressor is active when bound to a **corepressor**, often the end product of the pathway.



Part a. R N A polymerase bound at promoter. Repressor molecule is inactive and does not bind to operator (just downstream of promoter). Genes (downstream of operator) are expressed.

Part b. Genes not expressed. Active repressor with bound corepressors; binds to operator. R N A polymerase can't bind to promoter (upstream of operator); genes (downstream of operator) not expressed.

- In an inducible operon (usually encoding catabolic enzymes), binding of an **inducer** to an innately active repressor inactivates the repressor and turns on transcription.



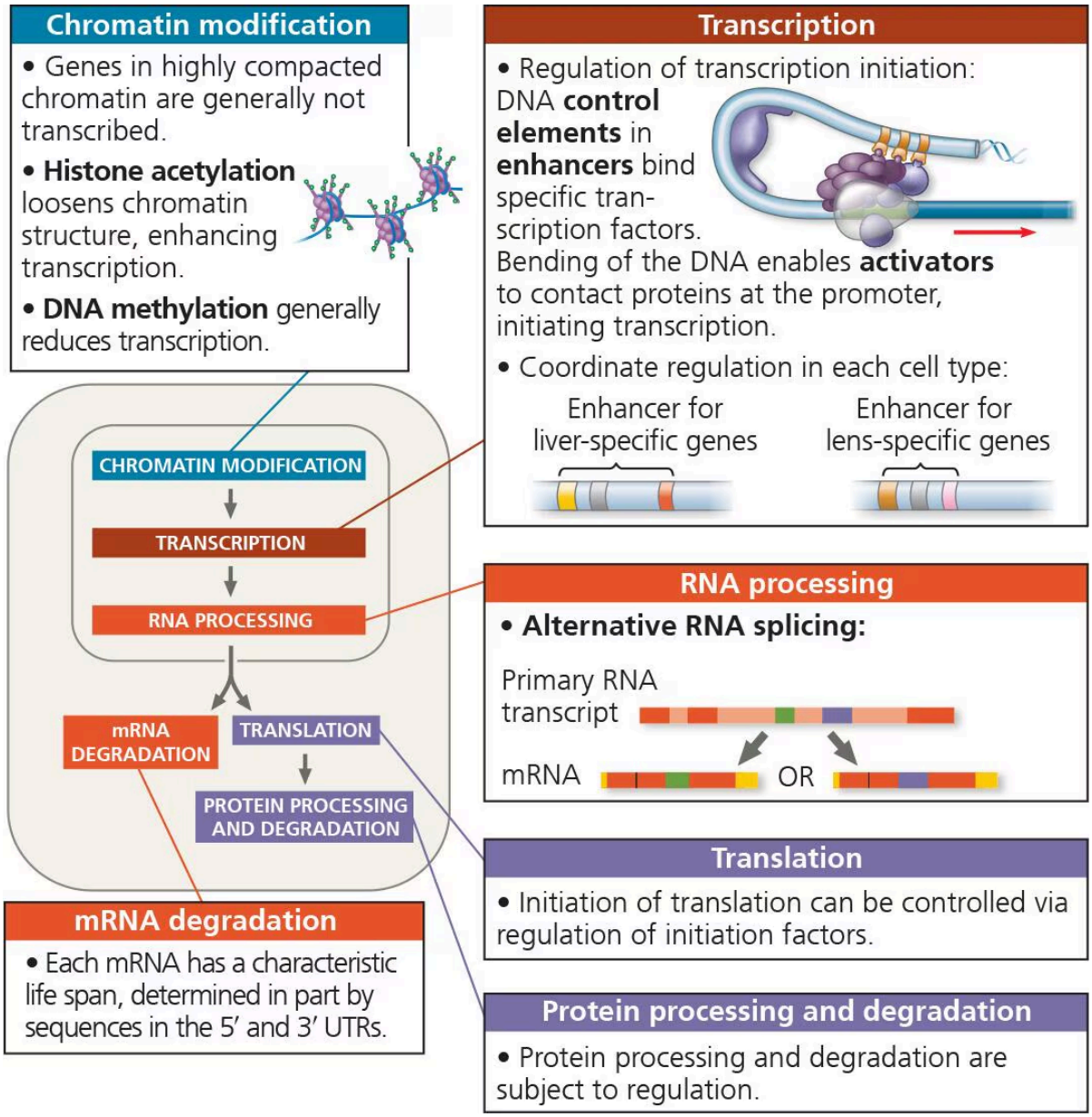
Part a. Genes not expressed. Repressor active; no inducer present. Binds to operator (downstream of promoter); R N A polymerase can not bind; genes not expressed.

Part b. Gene expressed. Inactive repressor with bound inducer. Does not bind to operator; R N A polymerase binds to promoter and genes are expressed.

- Some operons have positive gene regulation via a stimulatory **activator** protein (such as CRP, when activated by **cyclic AMP**), binds to a site within the promoter and stimulates transcription.

Compare and contrast the roles of a corepressor and an inducer in negative regulation of an operon.

Concept 18.2: Eukaryotic gene expression is regulated at many stages



Eukaryotic gene expression regulation at many stages:

1.Chromatin modification: Genes in highly compacted chromatin not generally transcribed. Histone acetylation: seems to loosen chromatin structure, enhancing transcription. D N A methylation generally reduces transcription.

2.Transcription: Regulation of transcription initiation: D N A control elements in enhancers bind specific transcription factors. Bending of the D N A enables activators to contact proteins at the promoter, initiating transcription. D N A-bending proteins and groups of mediator proteins shown. Coordinates transcription:

Enhancer (with control elements) for liver-specific genes shown; as well as enhancer for lens specific genes with different control elements.

3.R N A processing: AlteR N Ative R N A splicing. Primary R N A transcript splices to m R N A carrying different exons.

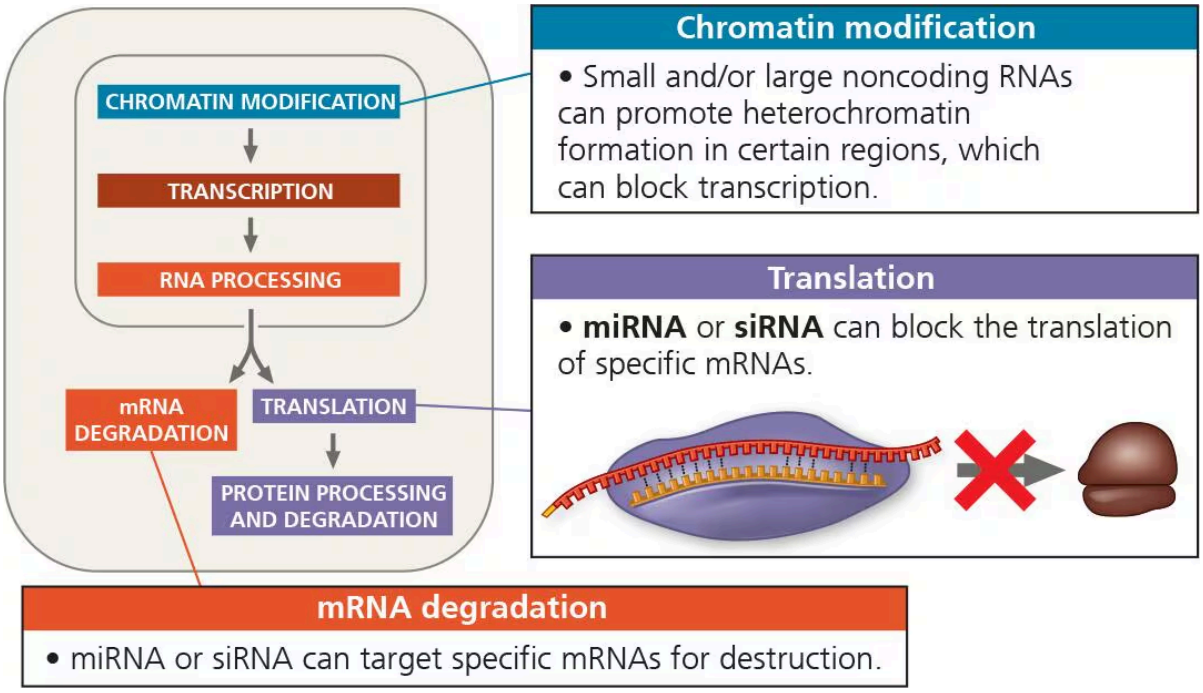
4.m R N A degradation: Each m R N A has a characteristic lifespan, determined in part by sequences in the 5 prime and 3 prime UTRs.

5.Translation: Initiation of translation can be controlled via regulation of initiation factors.

6.Protein processing and degradation: Protein processing and degradation are subject to regulation.

Describe what must happen in a cell for a gene specific to that cell type to be transcribed.

Concept 18.3: Noncoding RNAs play multiple roles in controlling gene expression



Non-coding R N As play multiple roles in controlling gene expression: chromatin modification, m R N A degradation, translation.

1.Chromatin modification: small and/or large noncoding R N As can promote heterochromatin formation in certain regions, which can block transcription.

2.Translation: miR N A or siR N A can block the translation of specific m R N As.

3.m R N A degradation: miR N A can target specific m R N As for destruction.

Why are miRNAs called noncoding RNAs? Explain how they participate in gene regulation.

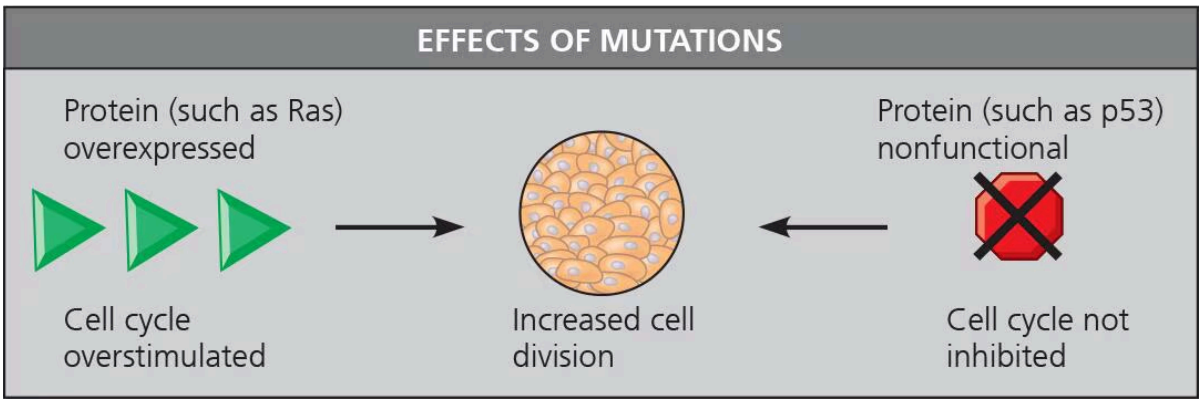
Concept 18.4: A program of differential gene expression leads to the different cell types in a multicellular organism

- Embryonic cells become committed to a certain fate (**determination**), and undergo **differentiation**, becoming specialized in structure and function for their determined fate. Cells differ in structure and function not because they contain different genomes but because they express different genes. **Morphogenesis** encompasses the processes that give shape to the organism and its various structures.
- Localized **cytoplasmic determinants** in the unfertilized egg are distributed differentially to daughter cells, where they regulate the expression of those cells' developmental fates. In the process called **induction**, signaling molecules from embryonic cells cause transcriptional changes in nearby target cells.
- Differentiation is marked by the appearance of tissue-specific proteins, which enable differentiated cells to carry out their specialized roles.
- In animals, **pattern formation**, the development of a spatial organization of tissues and organs, begins in the early embryo. **Positional information**, the molecular cues that control pattern formation, tells a cell its location relative to the body's axes and to other cells. In *Drosophila*, gradients of **morphogens** encoded by **maternal effect genes** determine the body axes. For example, the gradient of Bicoid protein determines the anterior-posterior axis.

Describe the two main processes that cause embryonic cells to head down different pathways to their final fates.

Concept 18.5: Cancer results from genetic changes that affect cell cycle control

- The products of **proto-oncogenes** and **tumor-suppressor genes** control cell division. A DNA change that makes a proto-oncogene overly active converts it to an **oncogene**, which may promote extra cell division and cancer. A tumor-suppressor gene encodes a protein that inhibits abnormal cell division. A mutation that reduces the activity of its protein product may lead to excessive cell division and cancer.
- Many proto-oncogenes and tumor-suppressor genes encode components of growth-stimulating and growth-inhibiting signaling pathways, respectively, and mutations in them can interfere with normal cell-signaling pathways. A hyperactive version of a protein in a stimulatory pathway, such as **Ras** (a G protein), functions as an oncogene protein. A defective version of a protein in an inhibitory pathway, such as p53 (a transcription activator), fails to act as a tumor suppressor.



Effects of mutations: cell cycle over stimulated; protein overexpressed; leads to increased cell division (mass of cells). Or, if cell cycle not inhibited, inhibitor protein absent, and cell division increased (mass of cells).

- In the multistep model of cancer development, normal cells are converted to cancer cells by the accumulation of mutations affecting proto-oncogenes and tumor-suppressor genes. Technical advances in DNA and mRNA sequencing are enabling cancer treatments that are more individually based.
- Genomics-based studies have resulted in researchers proposing four subtypes of breast cancer, based on expression of genes by tumor cells.
- Individuals who inherit a mutant allele of a proto-oncogene or tumor-suppressor gene have a predisposition to develop a particular cancer. Certain viruses promote cancer by integration of viral DNA into a cell's genome.

Compare the usual functions of proteins encoded by proto-oncogenes with those of proteins encoded by tumor-suppressor genes.